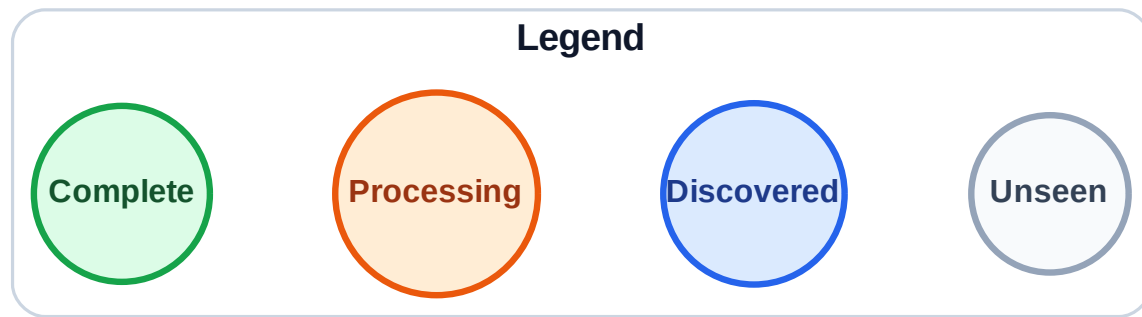
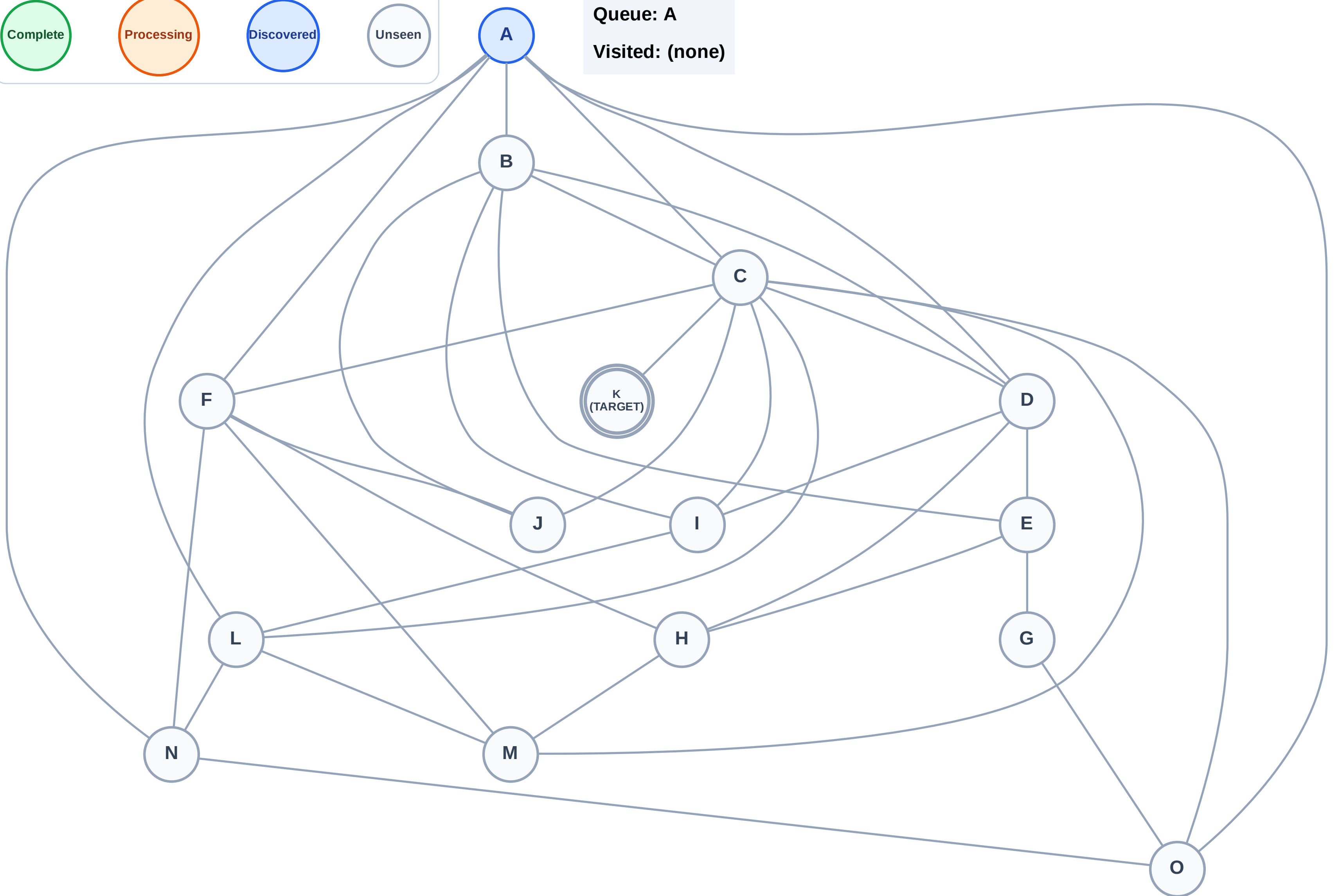


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

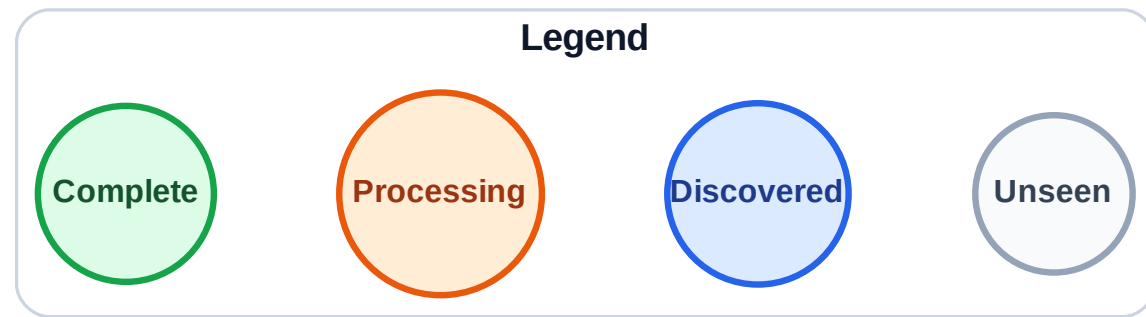


Queue: A
Visited: (none)



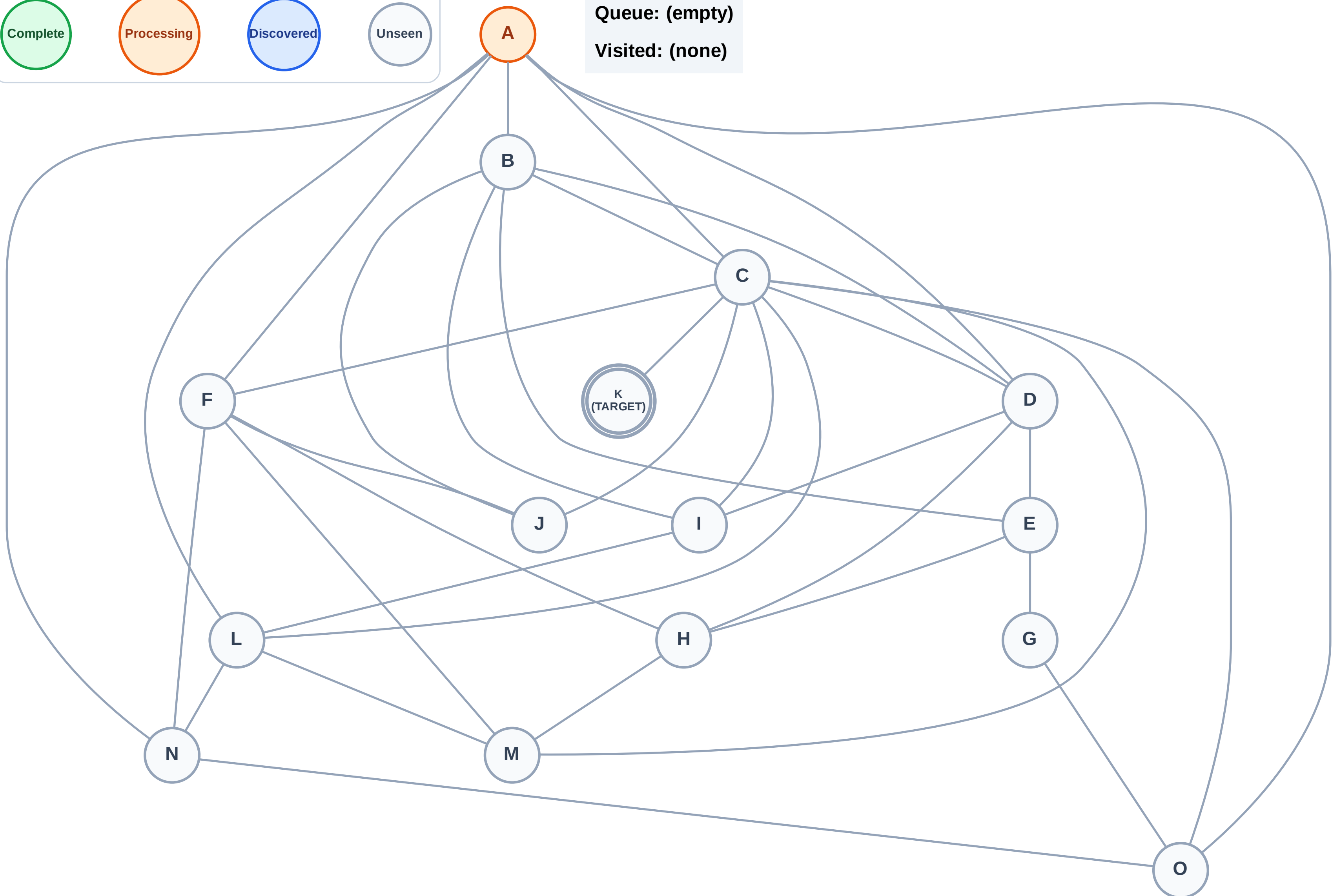
BFS Traversal

Step 2: Process node A



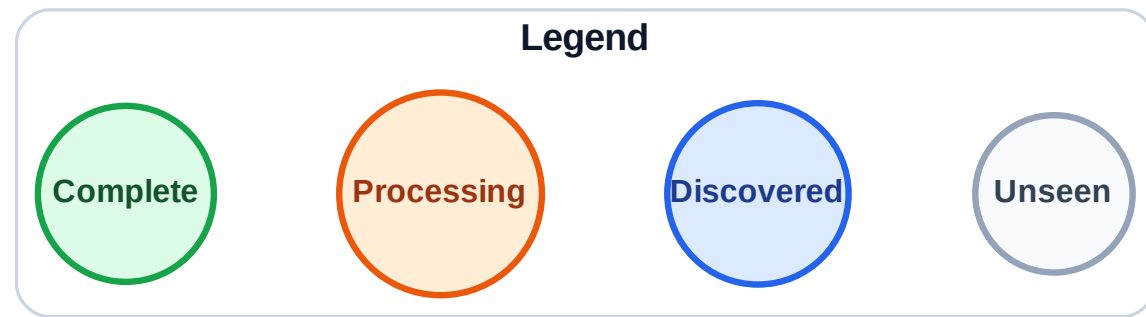
Queue: (empty)

Visited: (none)



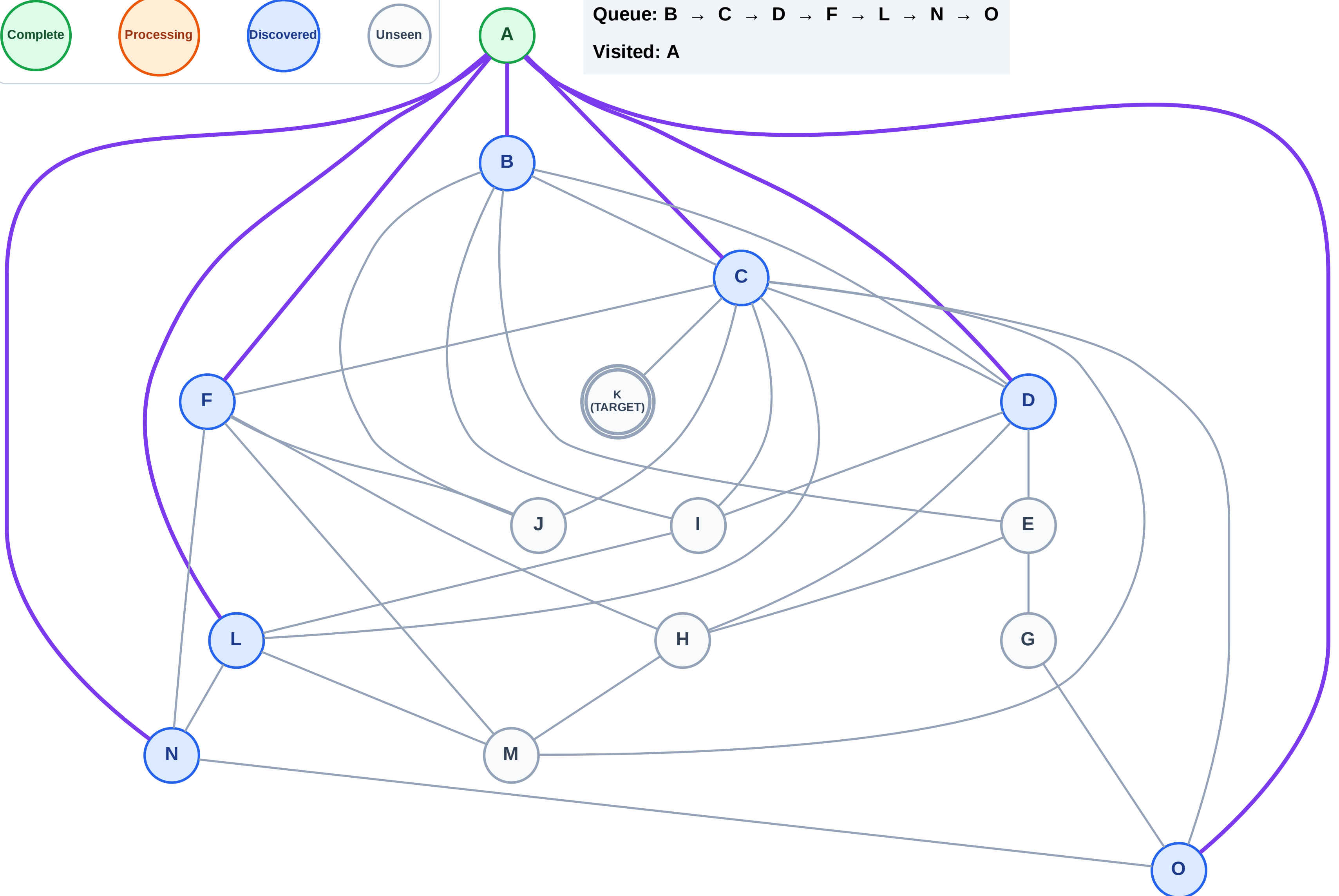
BFS Traversal

Step 3: Discover B, C, D, F, L, N, O from A



Queue: B → C → D → F → L → N → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

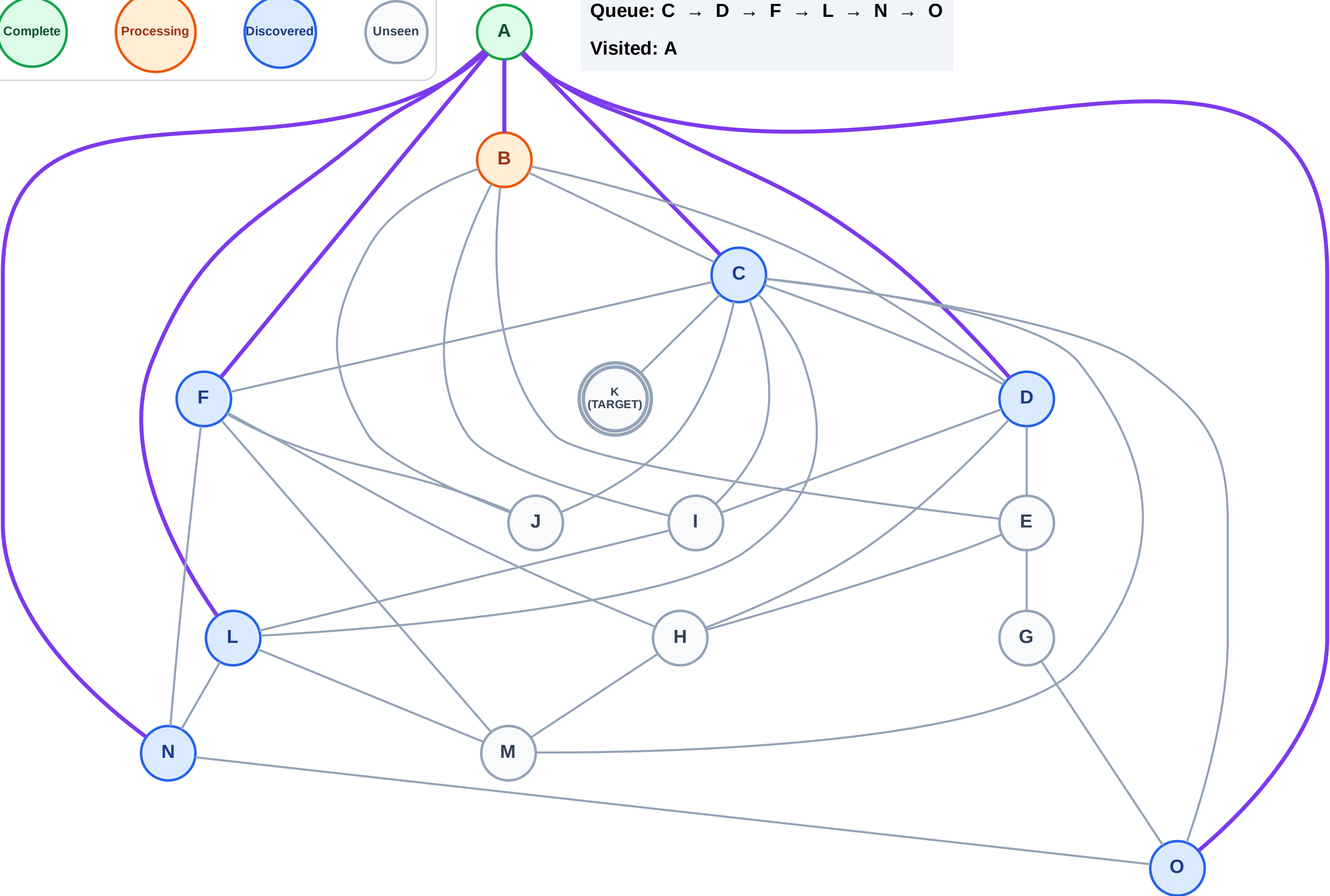
Processing

Discovered

Unseen

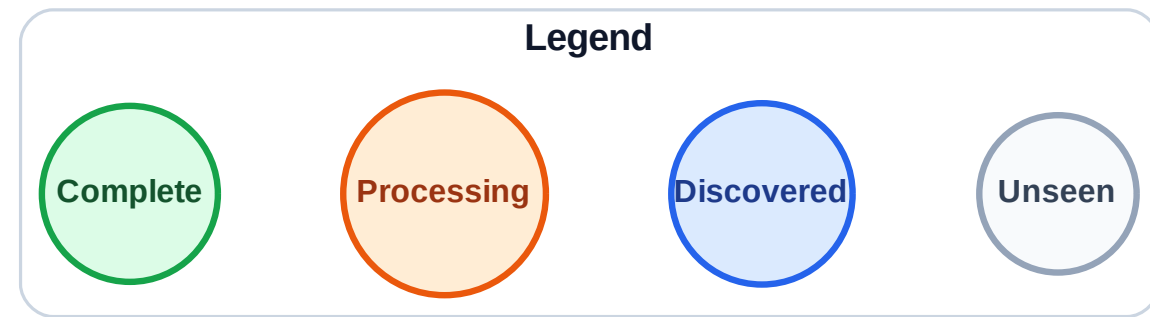
Queue: C → D → F → L → N → O

Visited: A



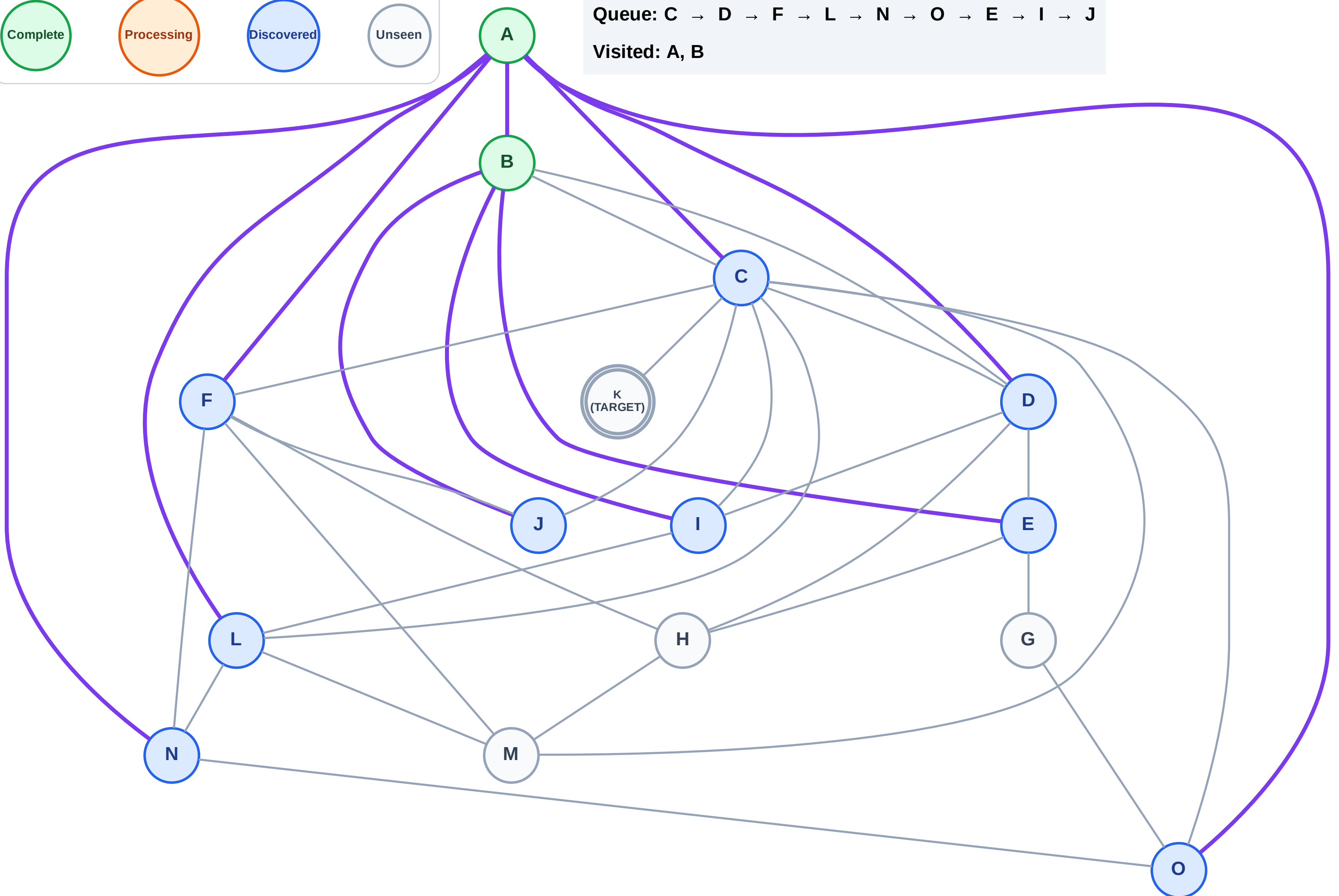
BFS Traversal

Step 5: Discover E, I, J from B



Queue: C → D → F → L → N → O → E → I → J

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

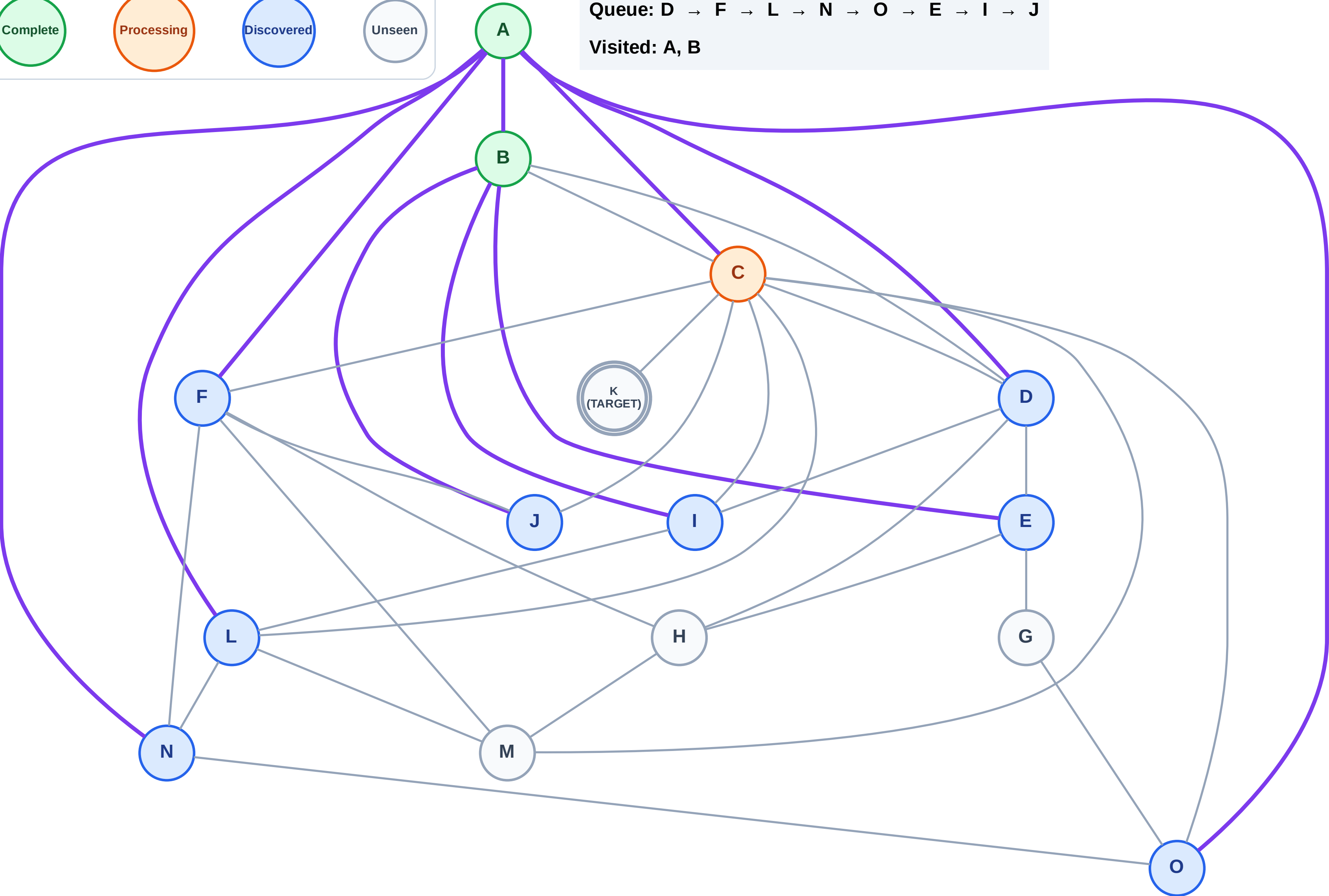
Processing

Discovered

Unseen

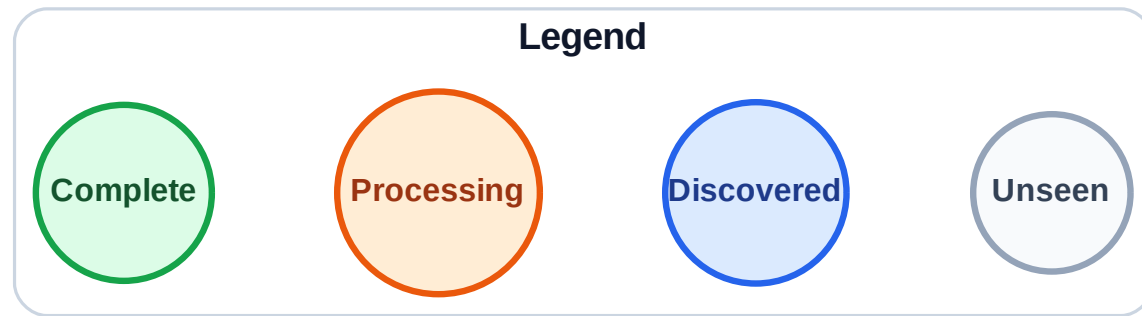
Queue: D → F → L → N → O → E → I → J

Visited: A, B



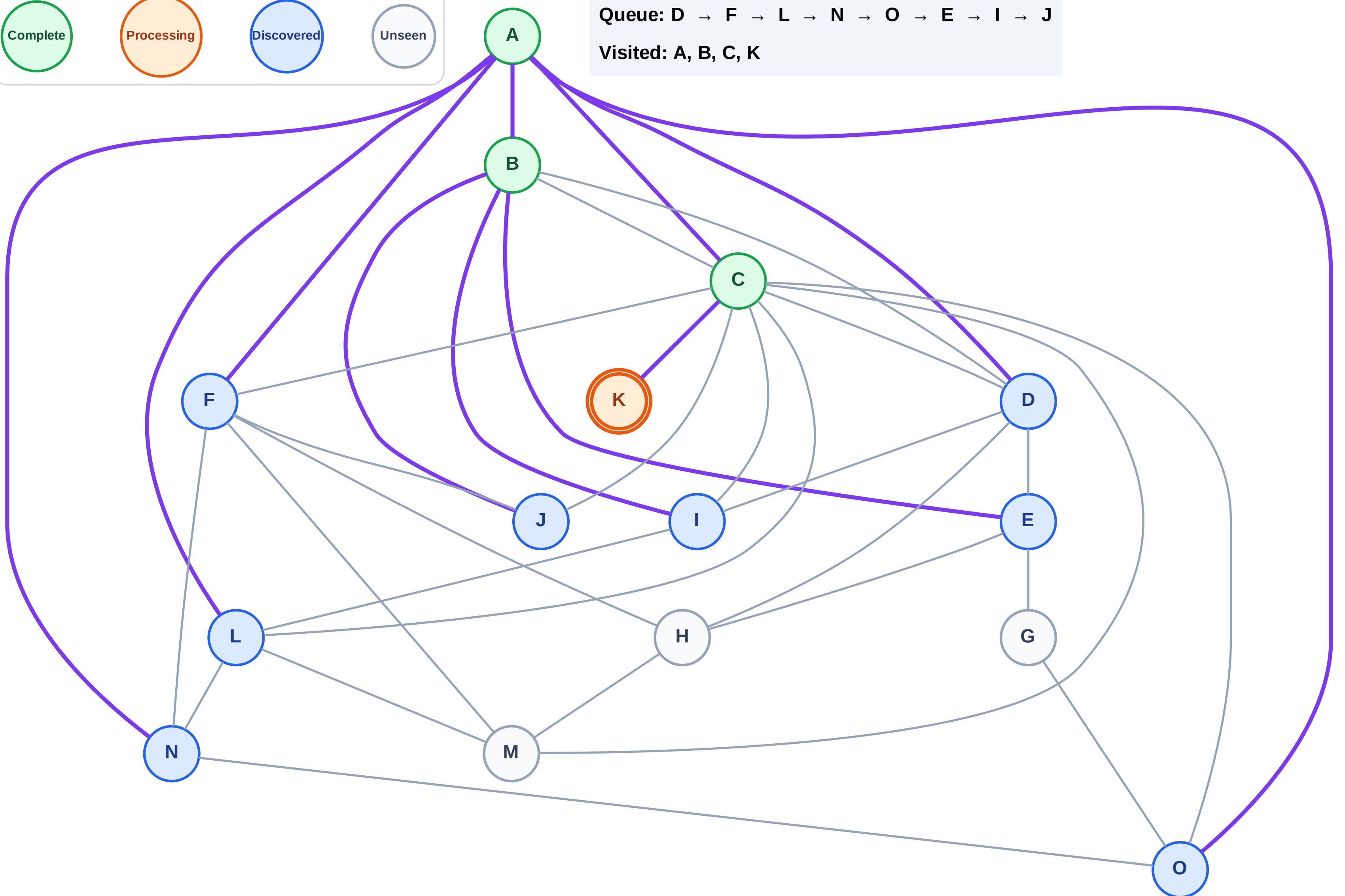
BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K



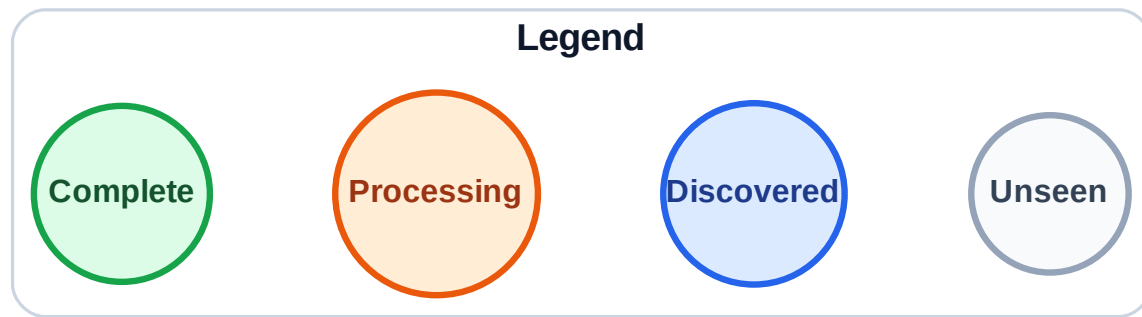
Queue: D → F → L → N → O → E → I → J

Visited: A, B, C, K

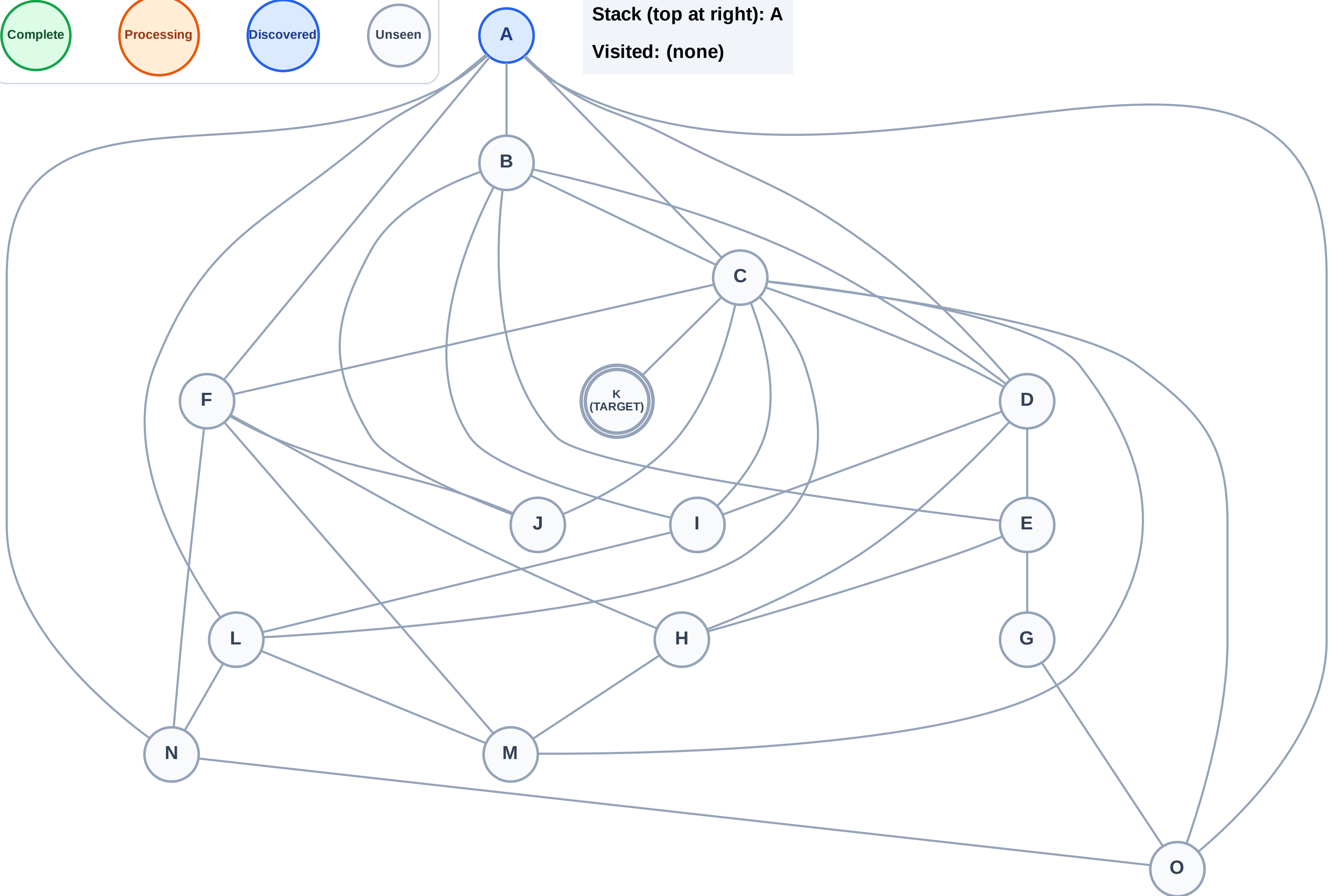


DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Stack (top at right): A
Visited: (none)



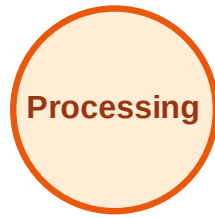
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



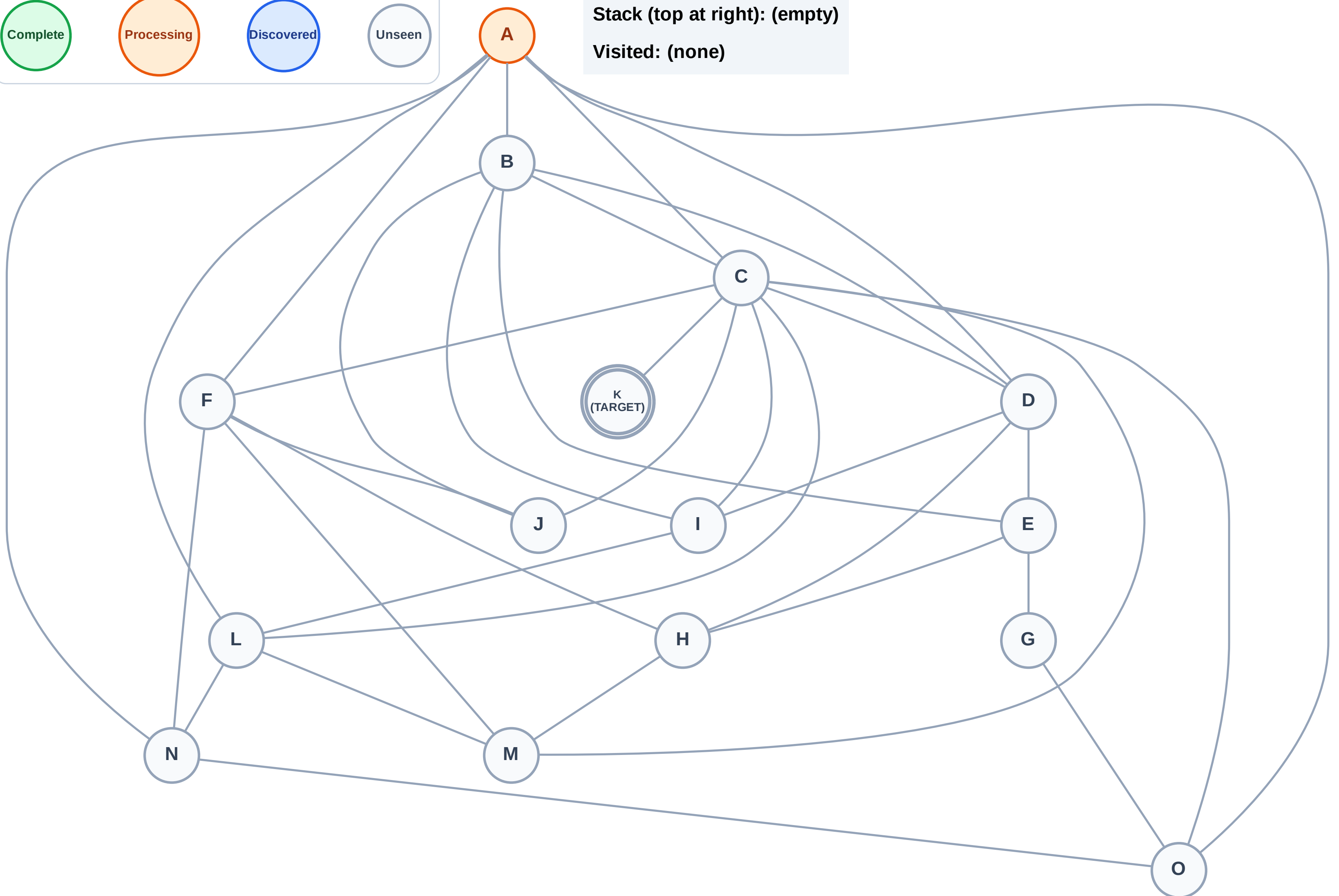
Discovered



Unseen

Stack (top at right): (empty)

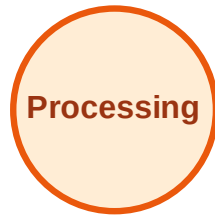
Visited: (none)



DFS Traversal

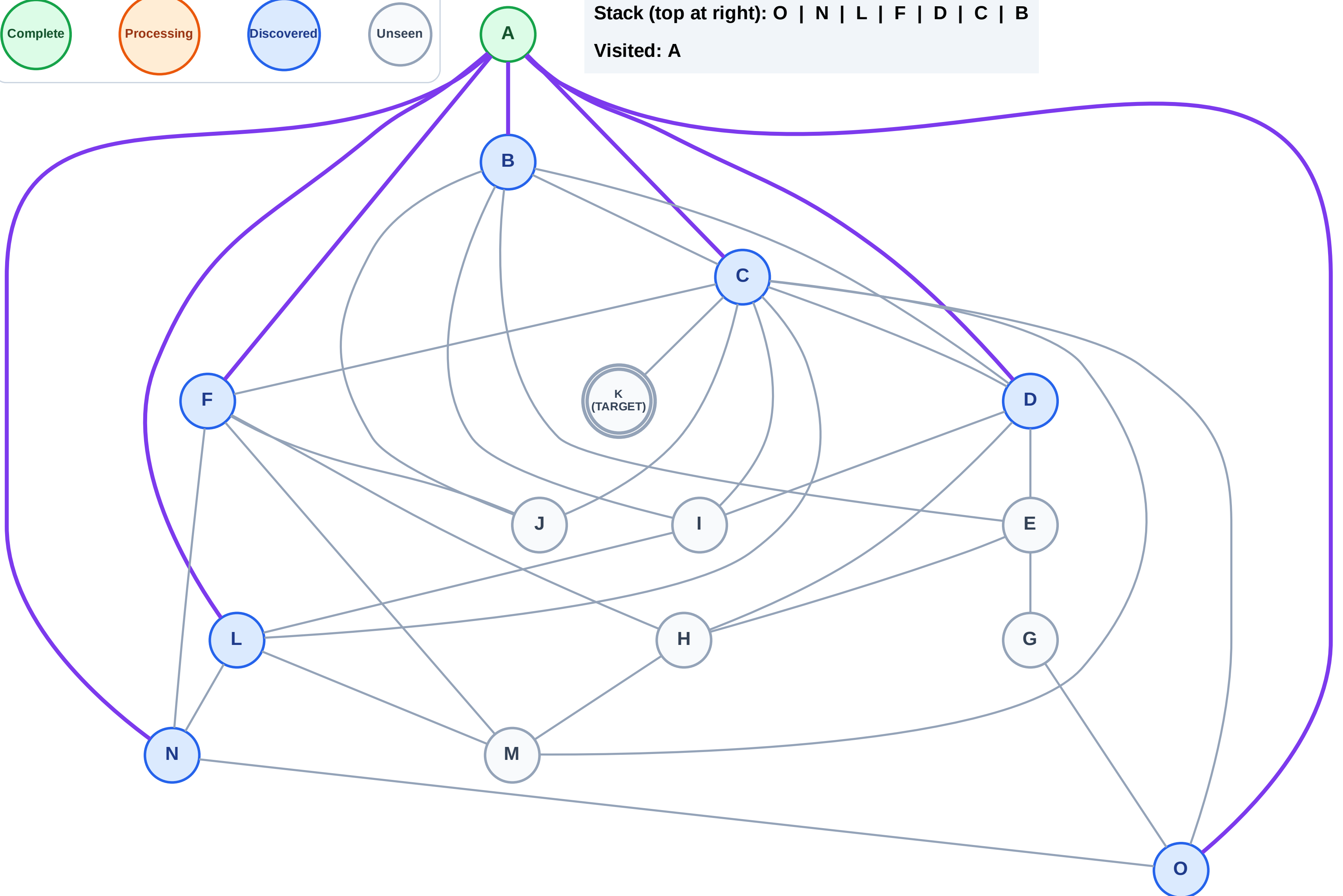
Step 3: Discover O, N, L, F, D, C, B from A

Legend



Stack (top at right): O | N | L | F | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

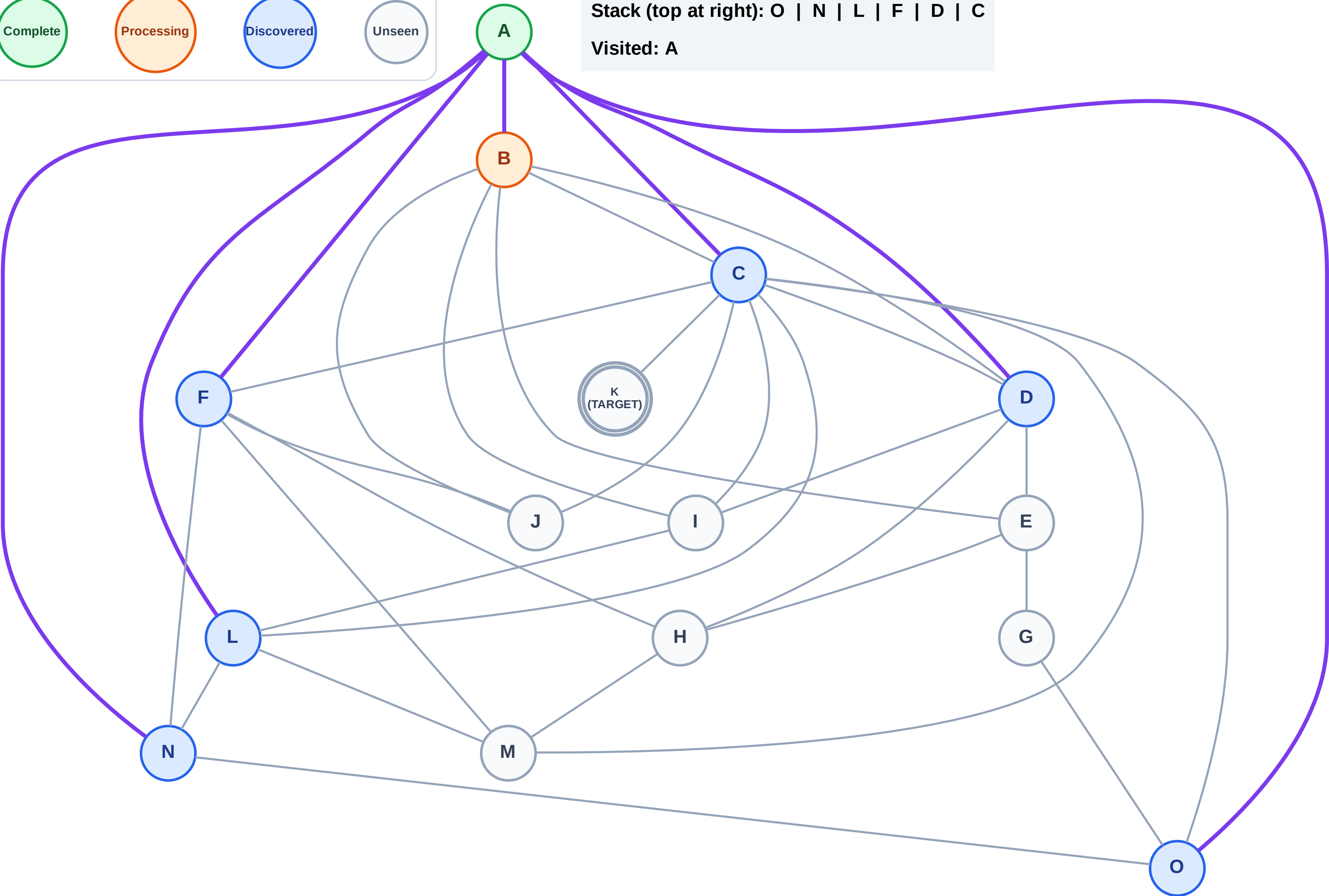
Processing

Discovered

Unseen

Stack (top at right): O | N | L | F | D | C

Visited: A



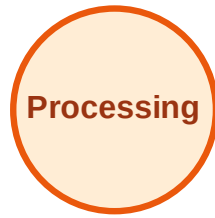
DFS Traversal

Step 5: Discover J, I, E from B

Legend



Complete



Processing



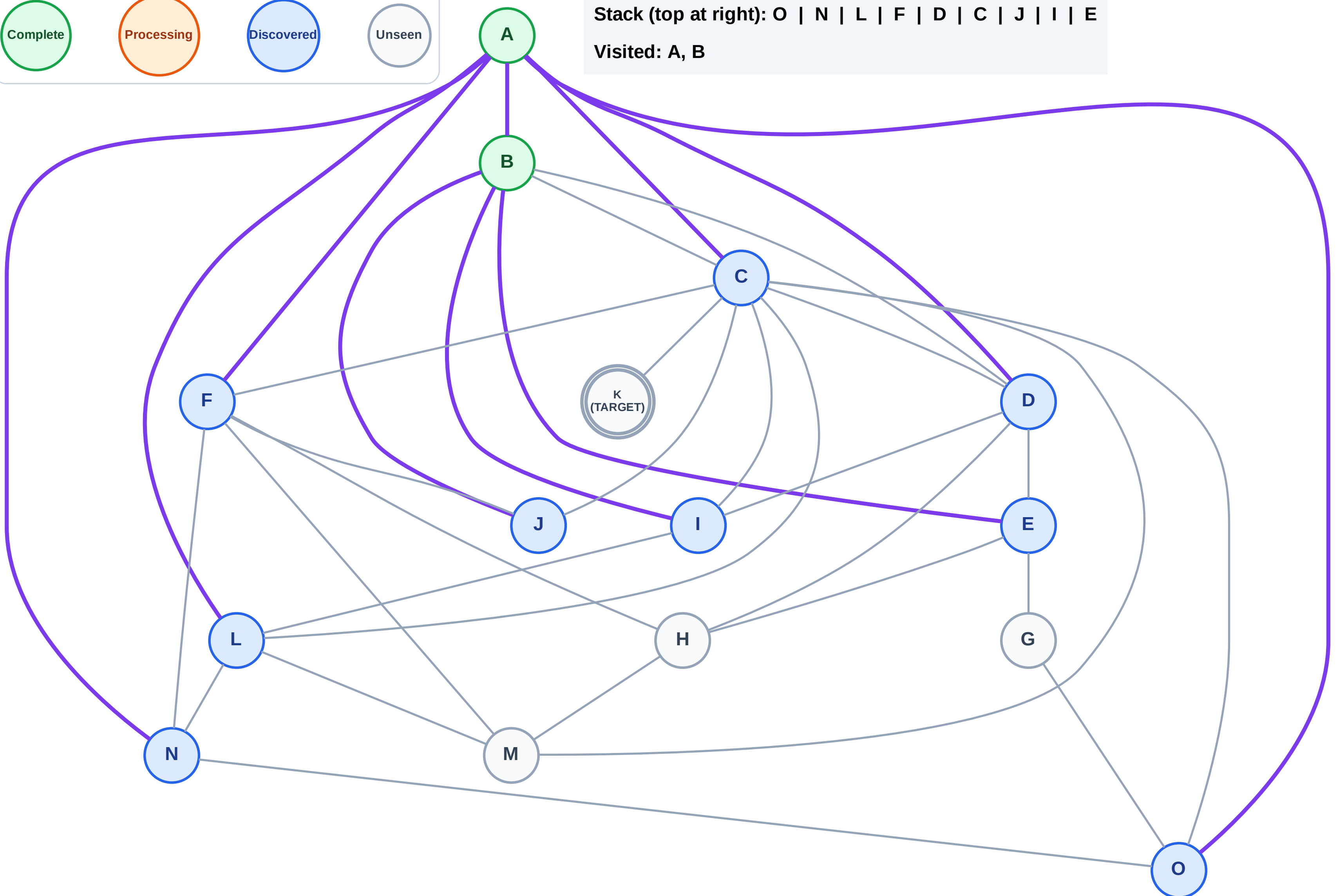
Discovered



Unseen

Stack (top at right): O | N | L | F | D | C | J | I | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

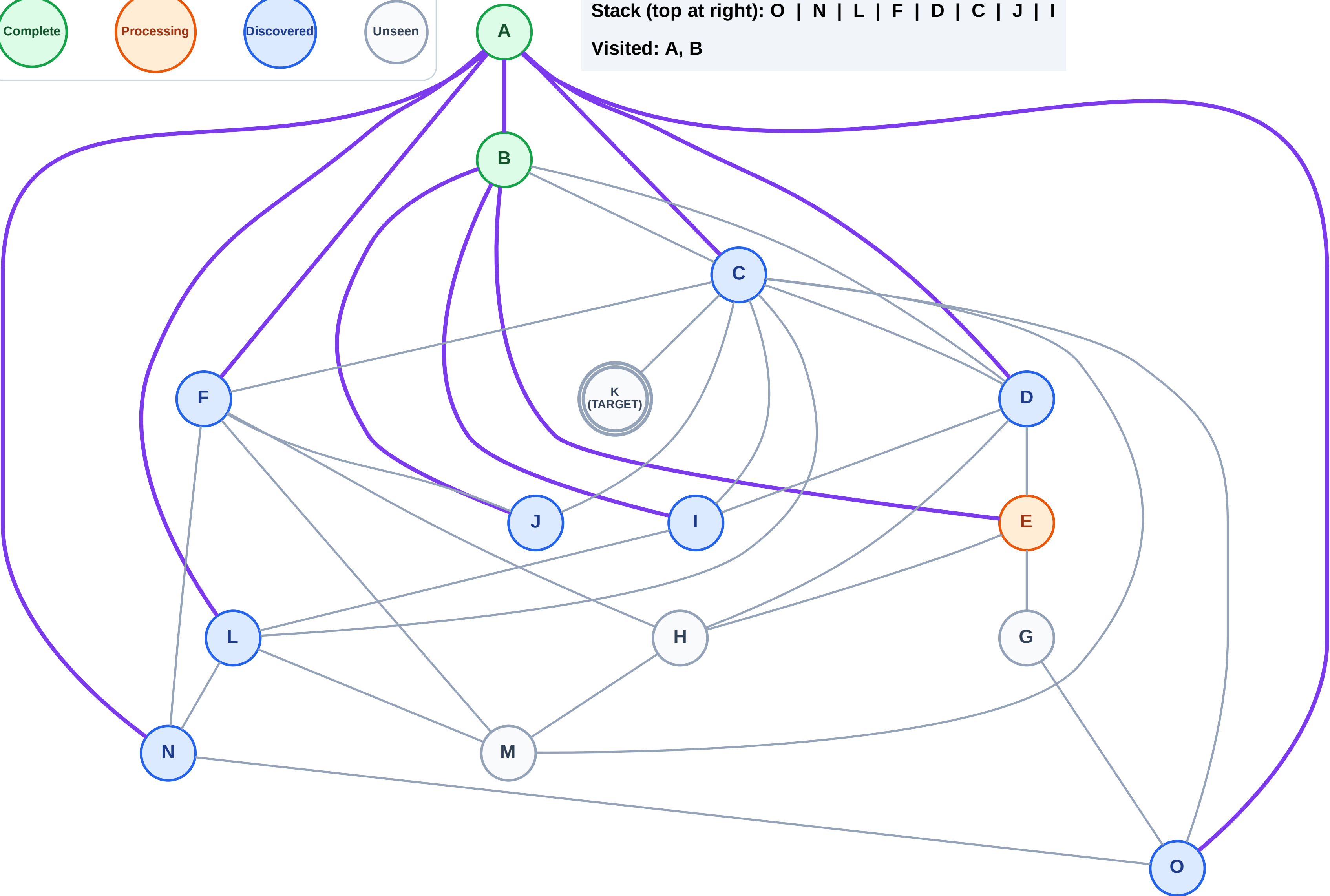
Processing

Discovered

Unseen

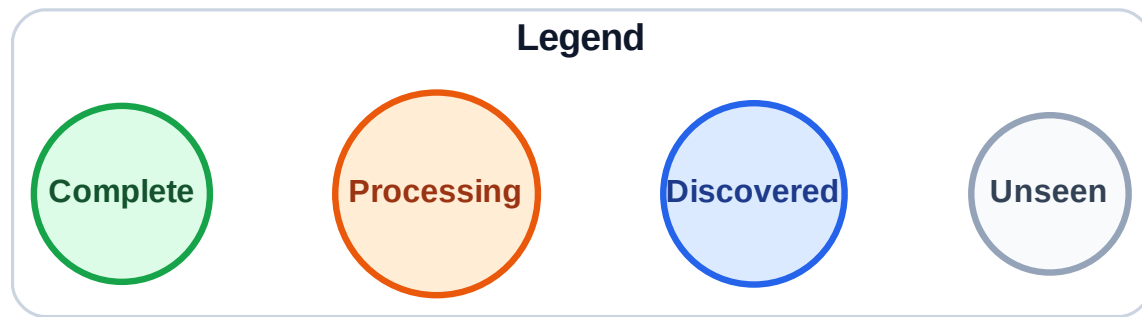
Stack (top at right): O | N | L | F | D | C | J | I

Visited: A, B



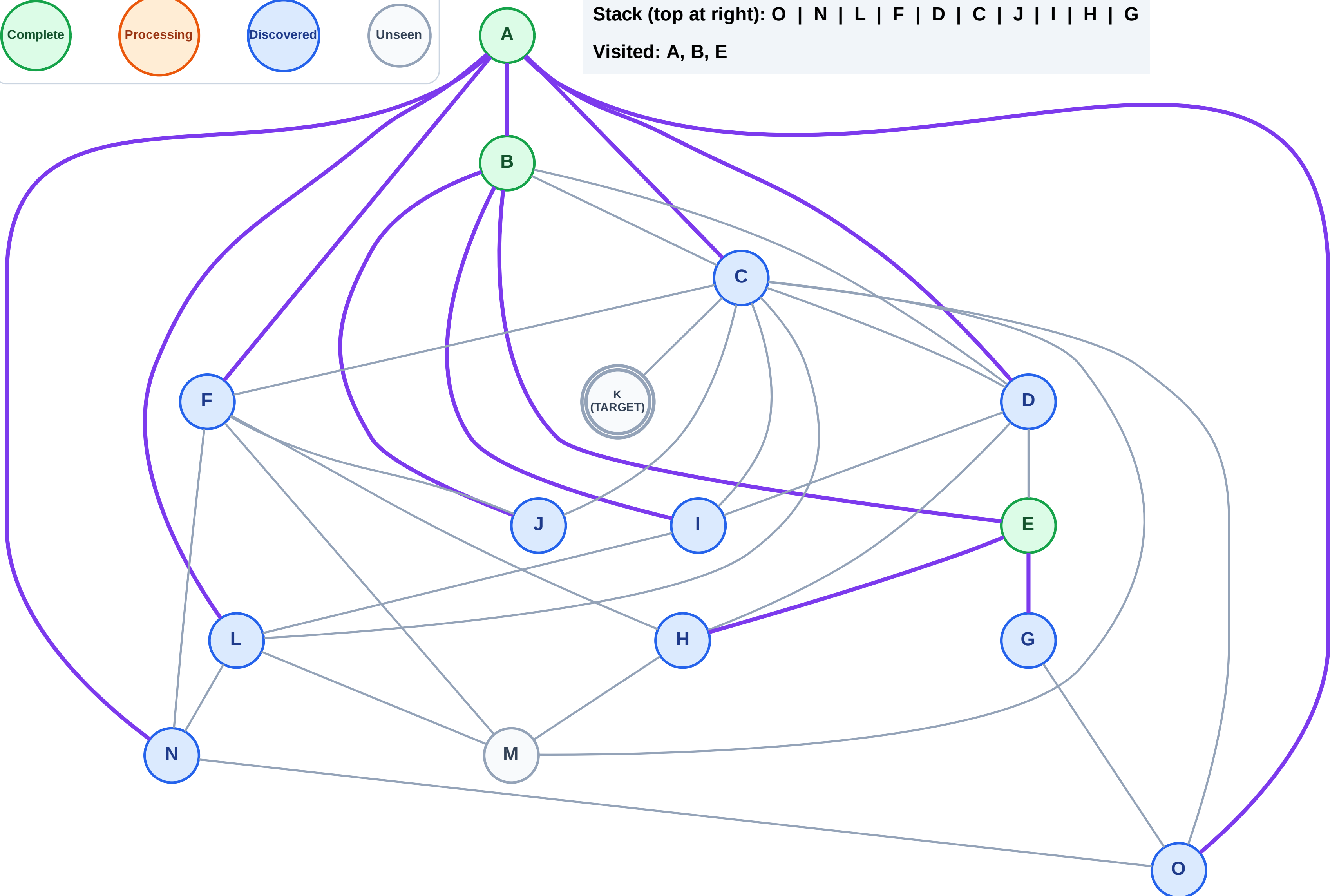
DFS Traversal

Step 7: Discover H, G from E



Stack (top at right): O | N | L | F | D | C | J | I | H | G

Visited: A, B, E



DFS Traversal

Step 8: Process node G

Legend

Complete

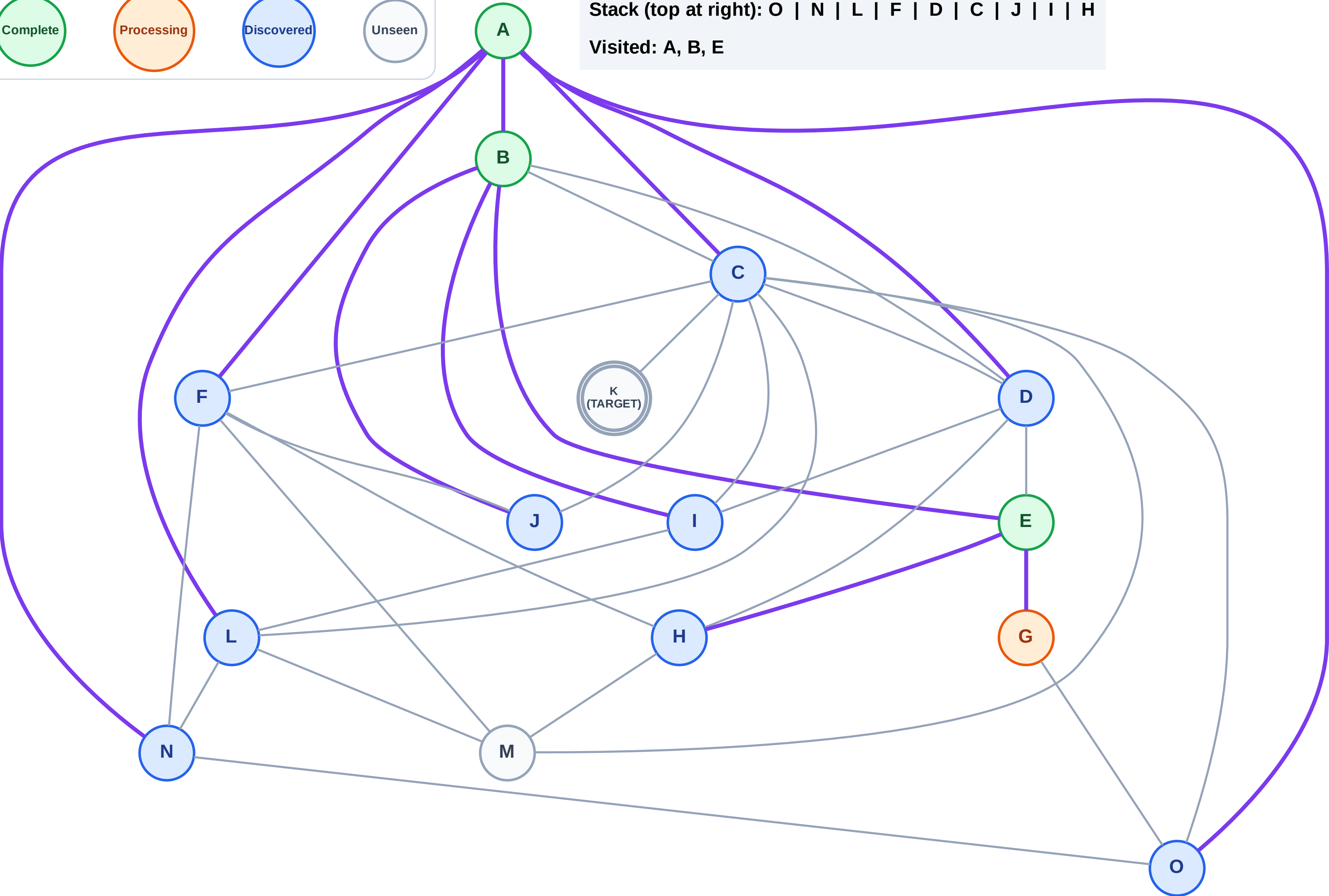
Processing

Discovered

Unseen

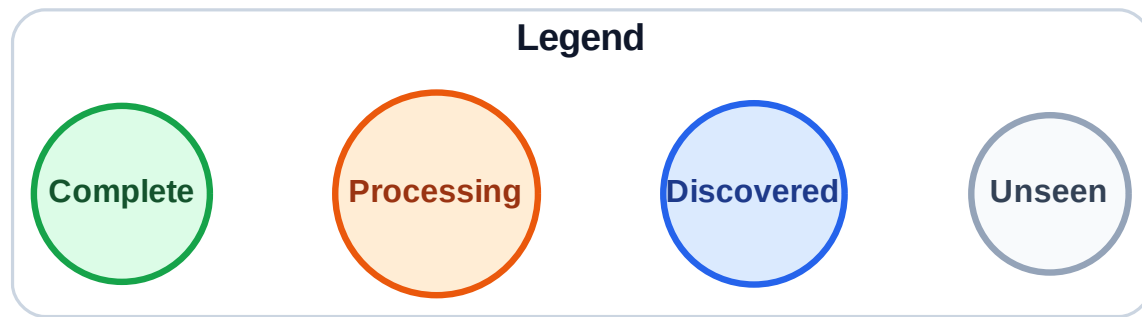
Stack (top at right): O | N | L | F | D | C | J | I | H

Visited: A, B, E



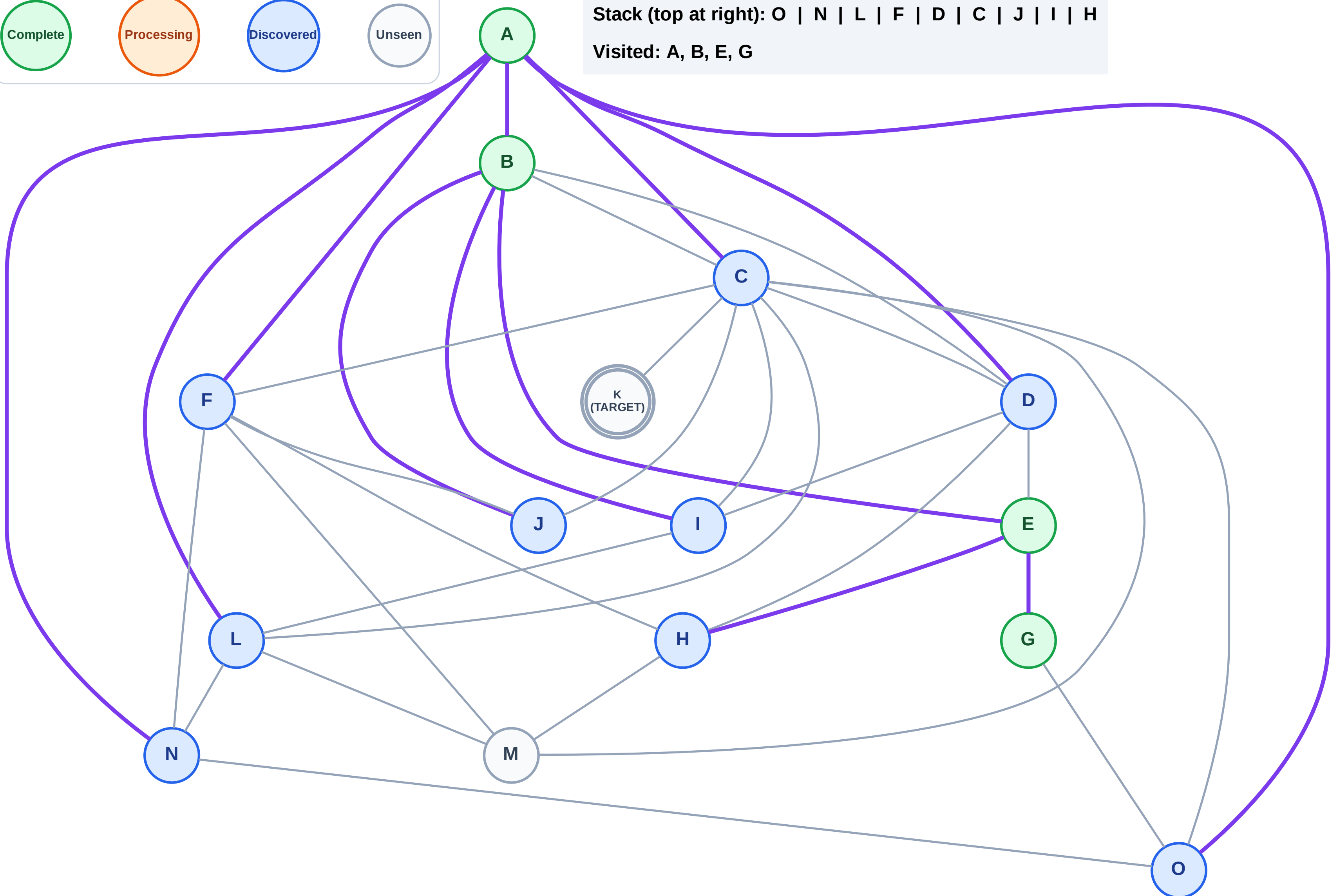
DFS Traversal

Step 9: No new neighbors from G



Stack (top at right): O | N | L | F | D | C | J | I | H

Visited: A, B, E, G



DFS Traversal

Step 10: Process node H

Legend

Complete

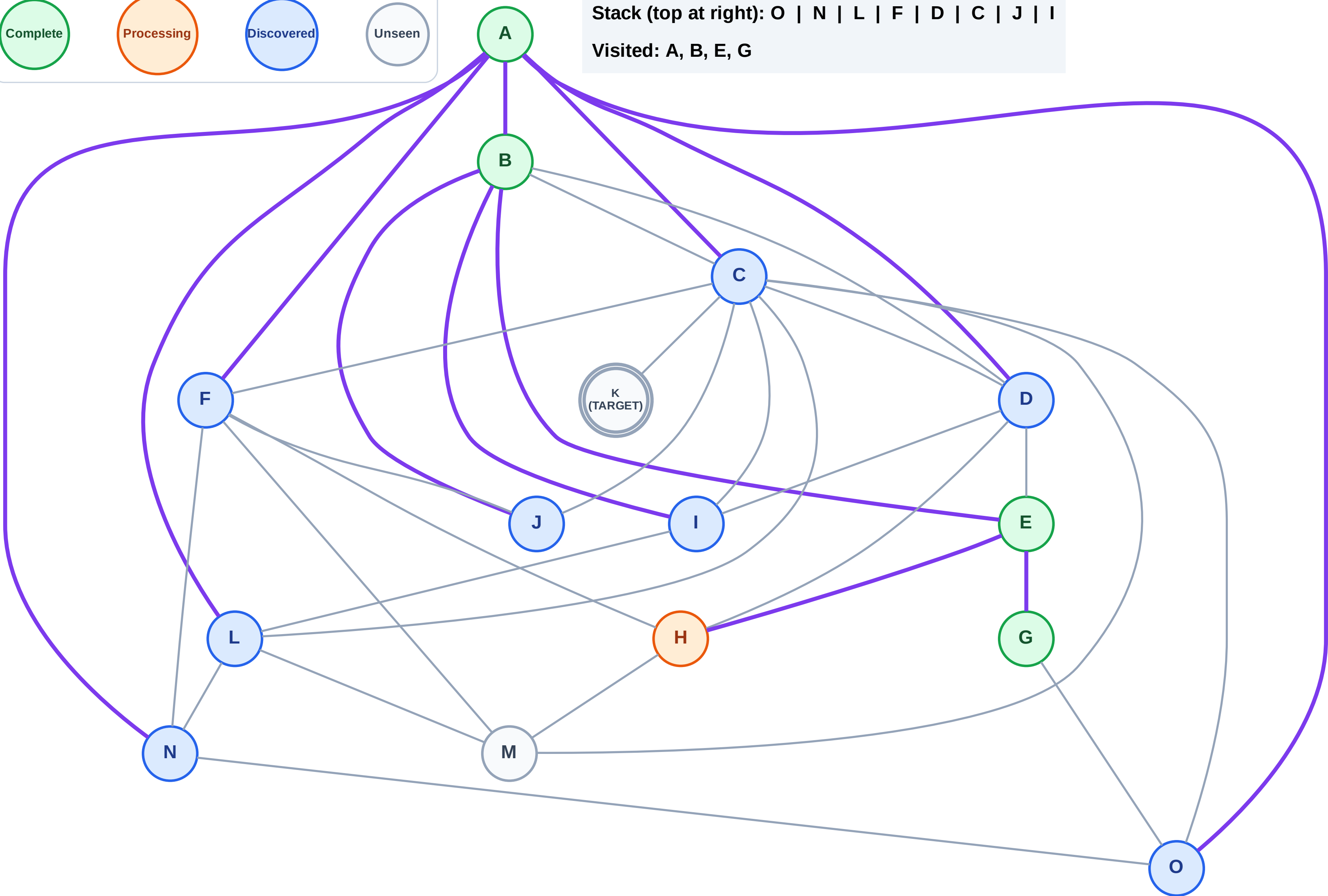
Processing

Discovered

Unseen

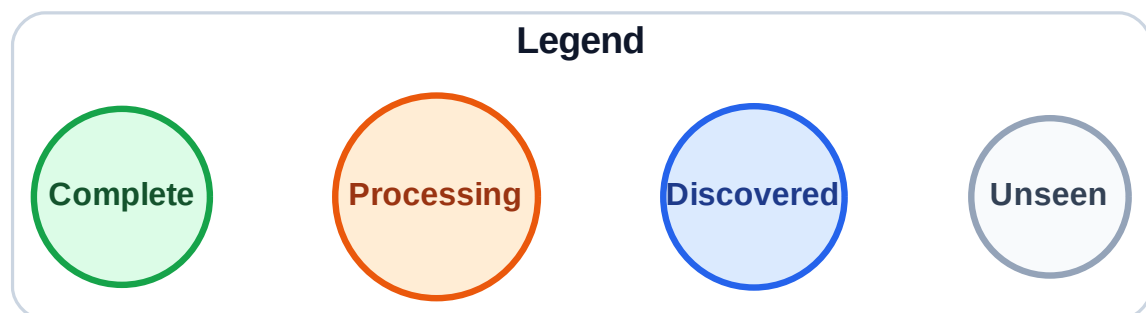
Stack (top at right): O | N | L | F | D | C | J | I

Visited: A, B, E, G



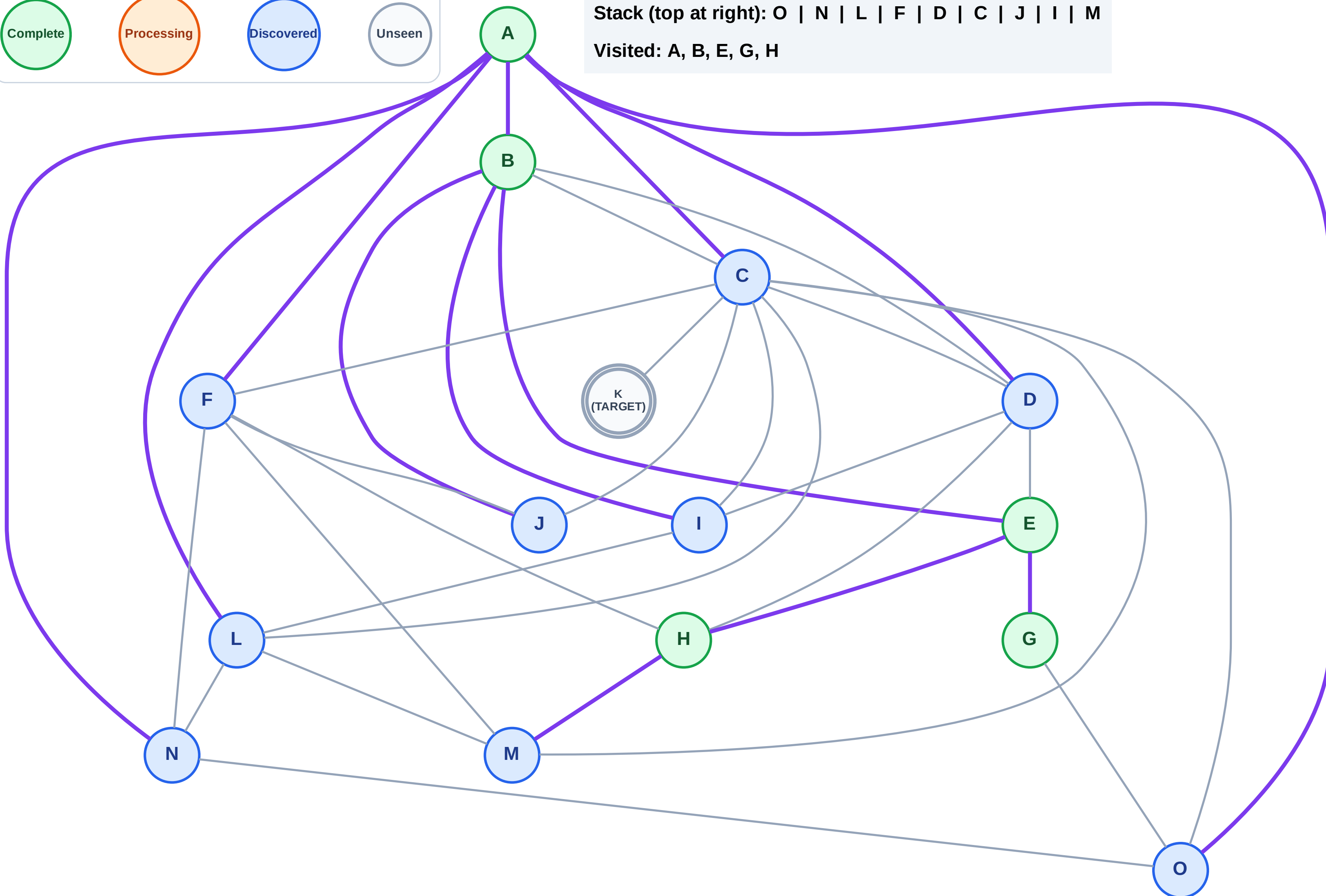
DFS Traversal

Step 11: Discover M from H



Stack (top at right): O | N | L | F | D | C | J | I | M

Visited: A, B, E, G, H



DFS Traversal

Step 12: Process node M

Legend

Complete

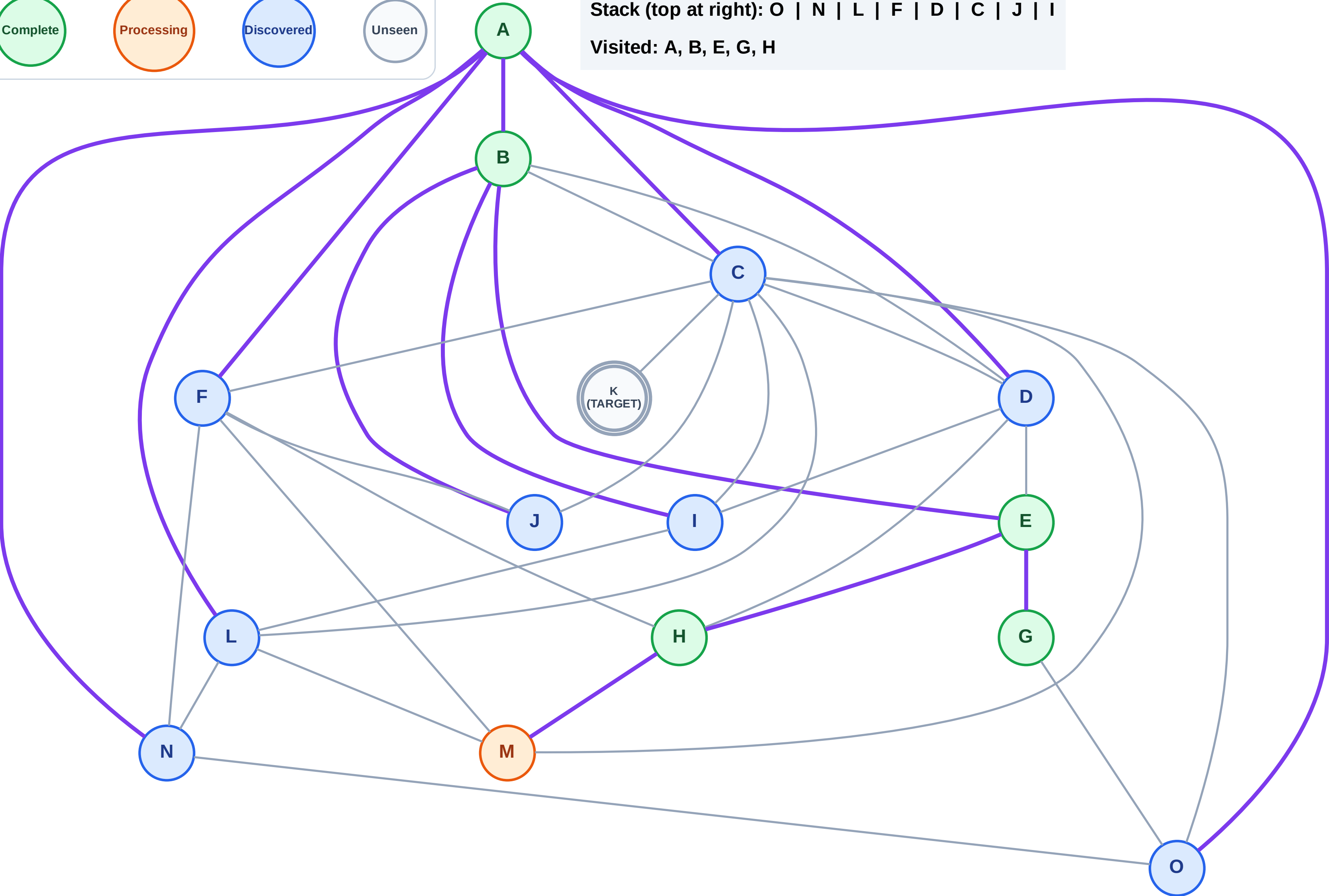
Processing

Discovered

Unseen

Stack (top at right): O | N | L | F | D | C | J | I

Visited: A, B, E, G, H



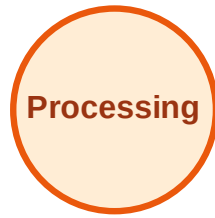
DFS Traversal

Step 13: No new neighbors from M

Legend



Complete



Processing



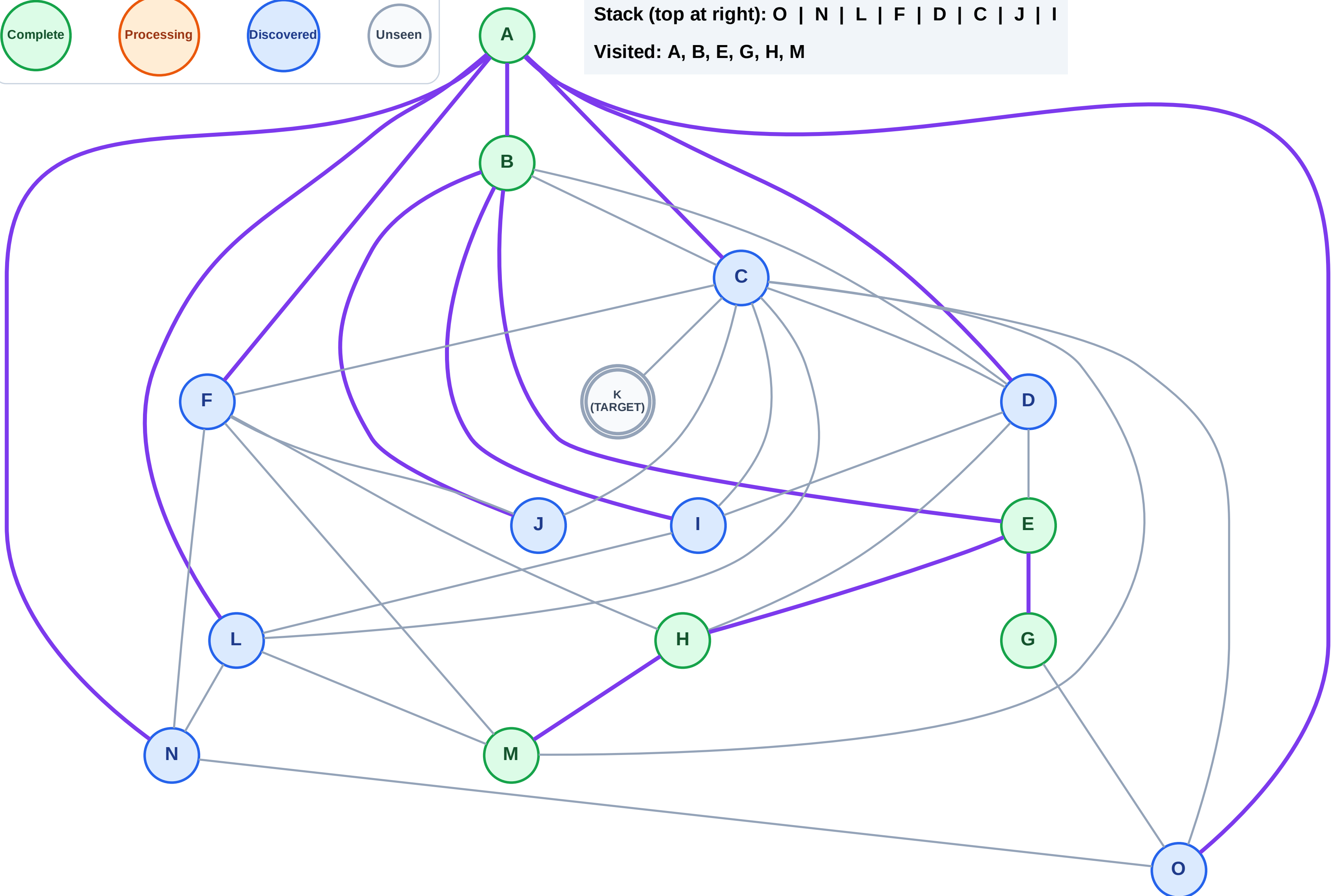
Discovered



Unseen

Stack (top at right): O | N | L | F | D | C | J | I

Visited: A, B, E, G, H, M



DFS Traversal

Step 14: Process node I

Legend

Complete

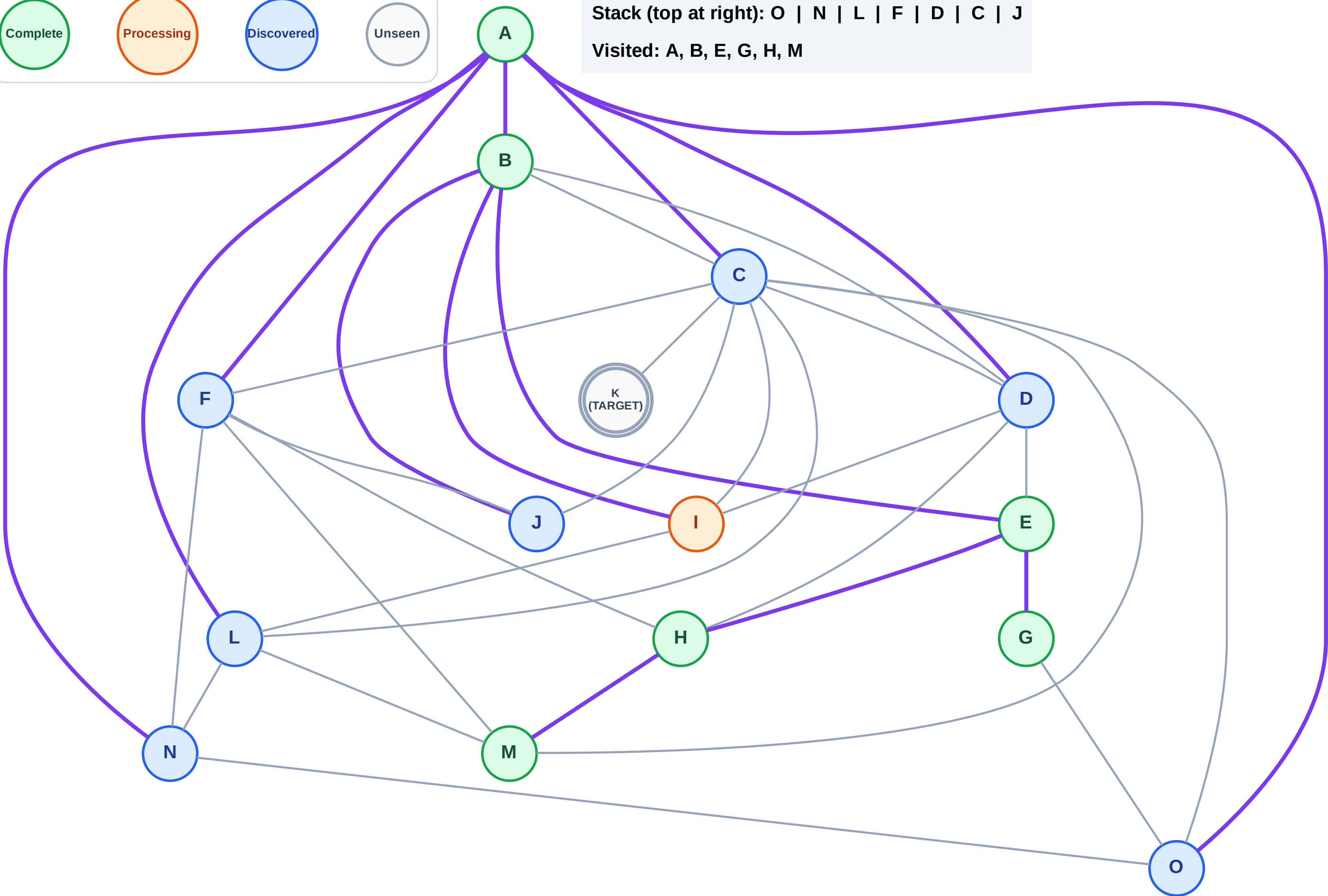
Processing

Discovered

Unseen

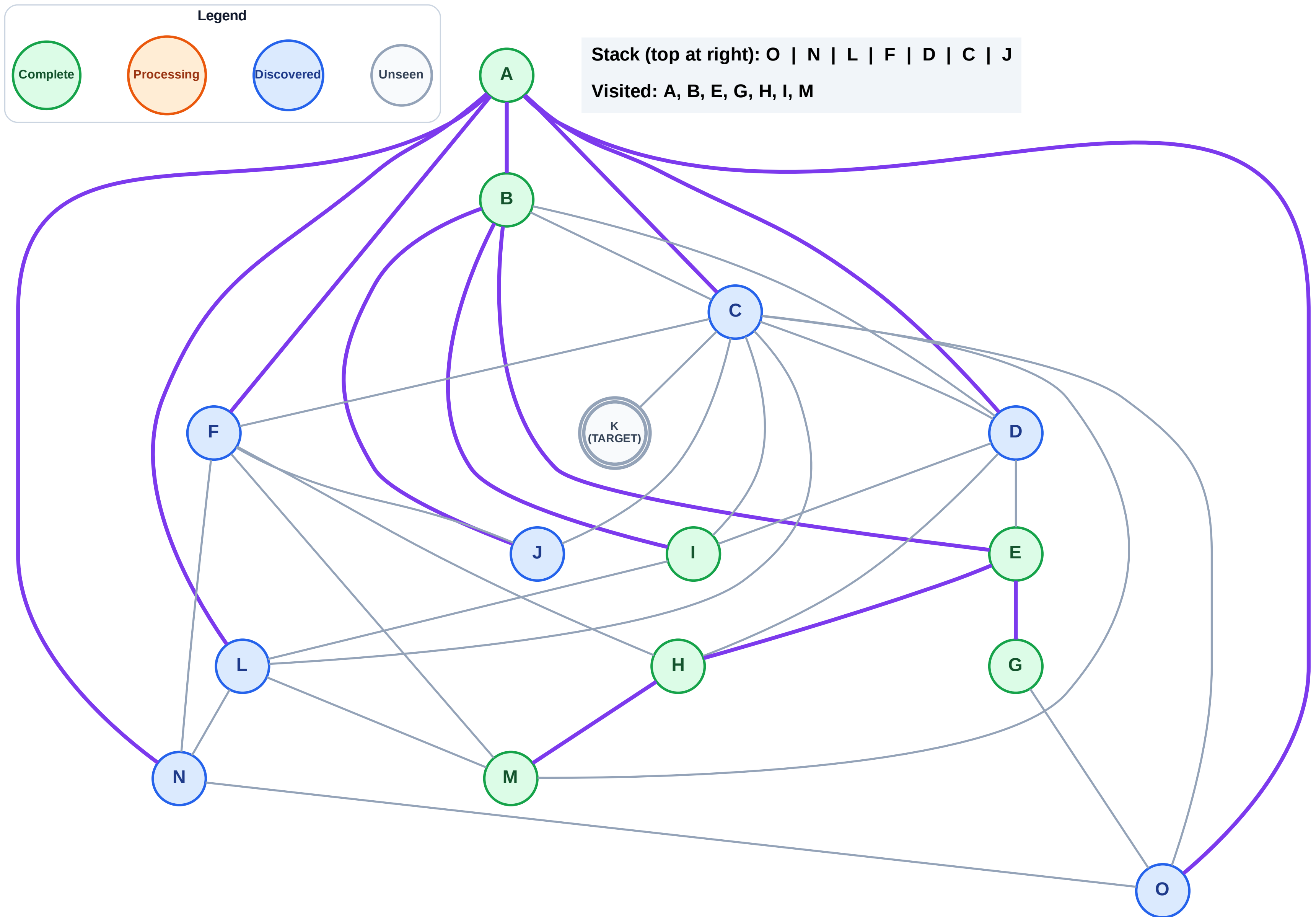
Stack (top at right): O | N | L | F | D | C | J

Visited: A, B, E, G, H, M



Step 15: No new neighbors from I

Step 15: No new neighbors from I



DFS Traversal

Step 16: Process node J

Legend

Complete

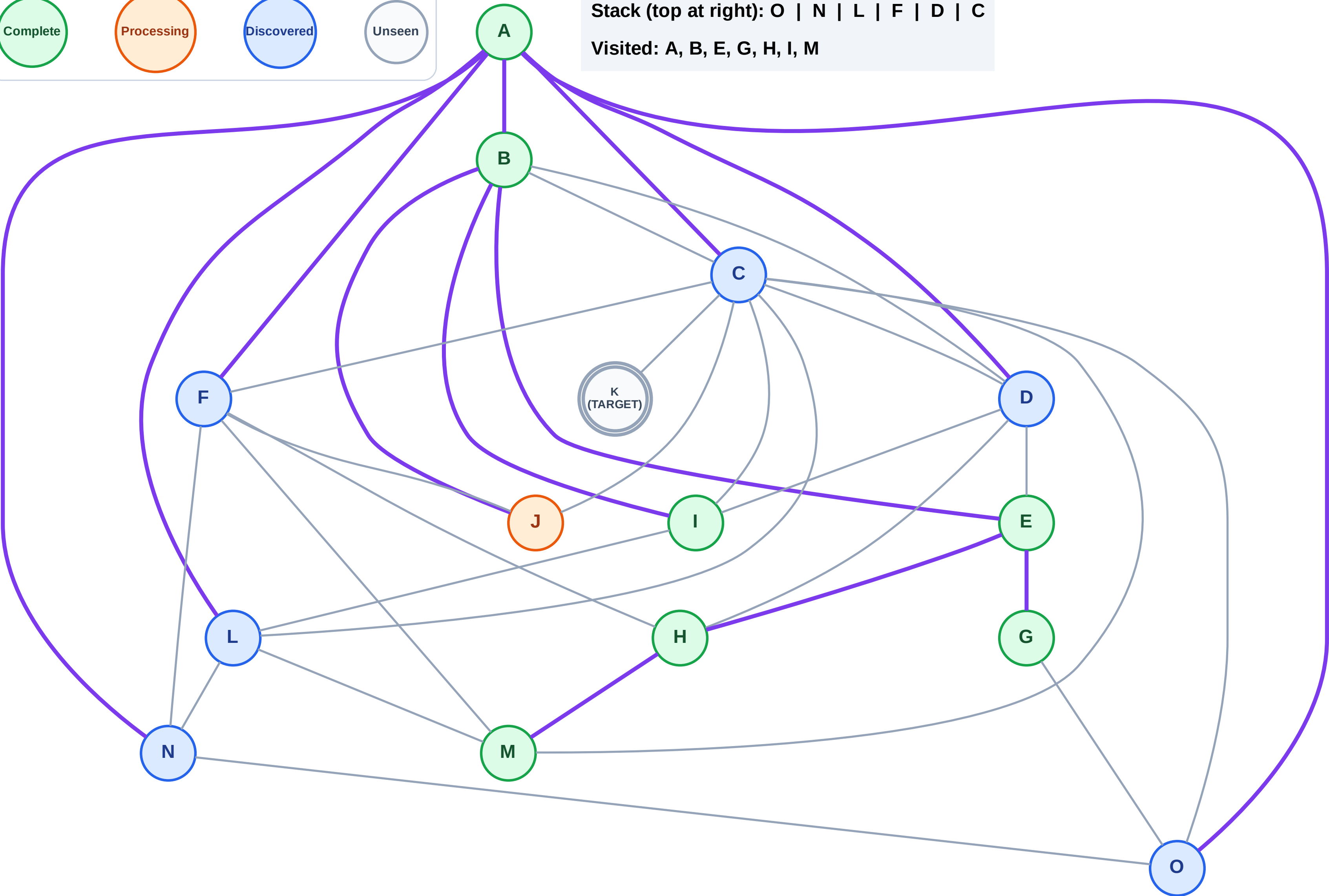
Processing

Discovered

Unseen

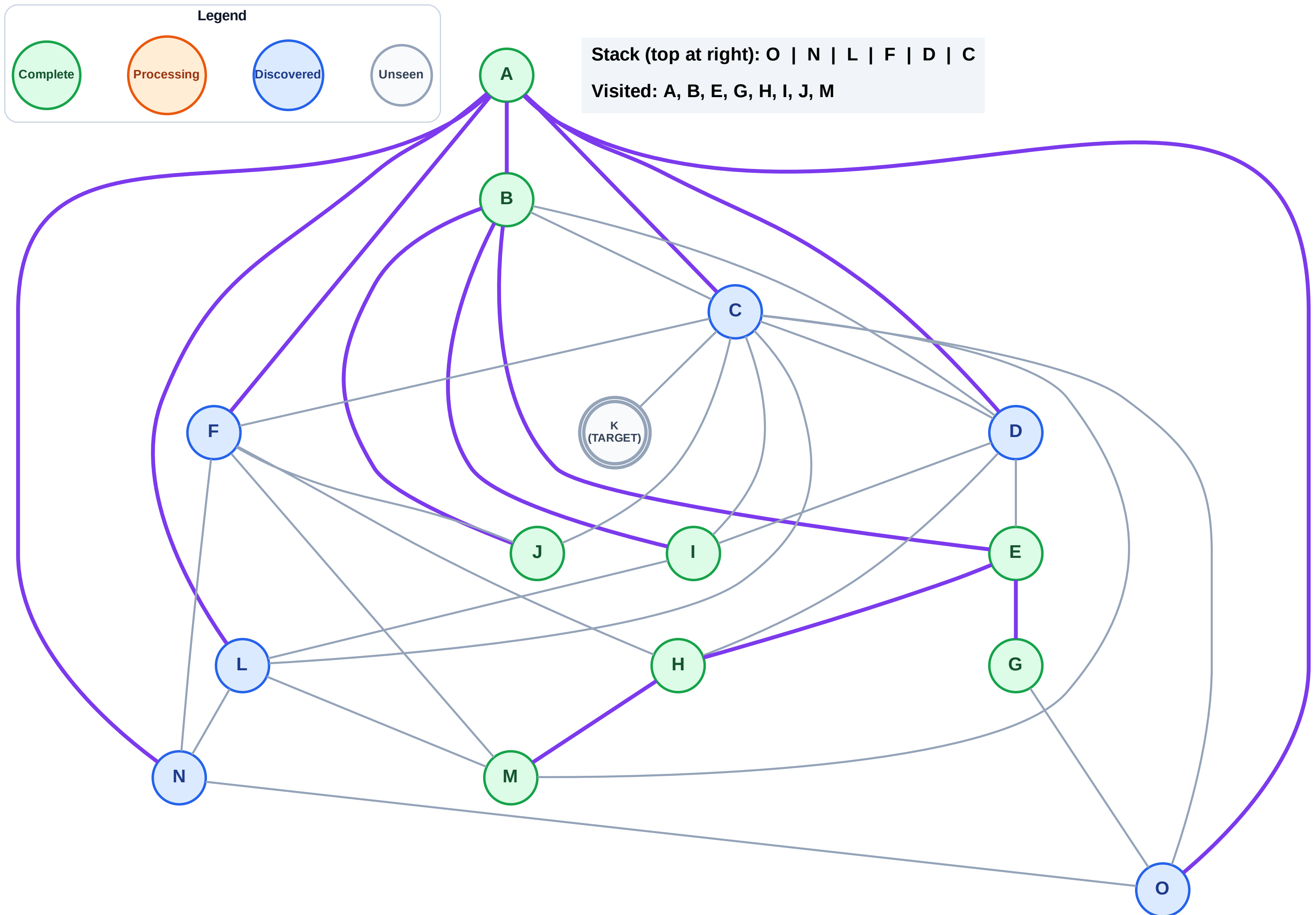
Stack (top at right): O | N | L | F | D | C

Visited: A, B, E, G, H, I, M



Step 17: No new neighbors from J

Step 17: No new neighbors from J



DFS Traversal

Step 18: Process node C

Legend

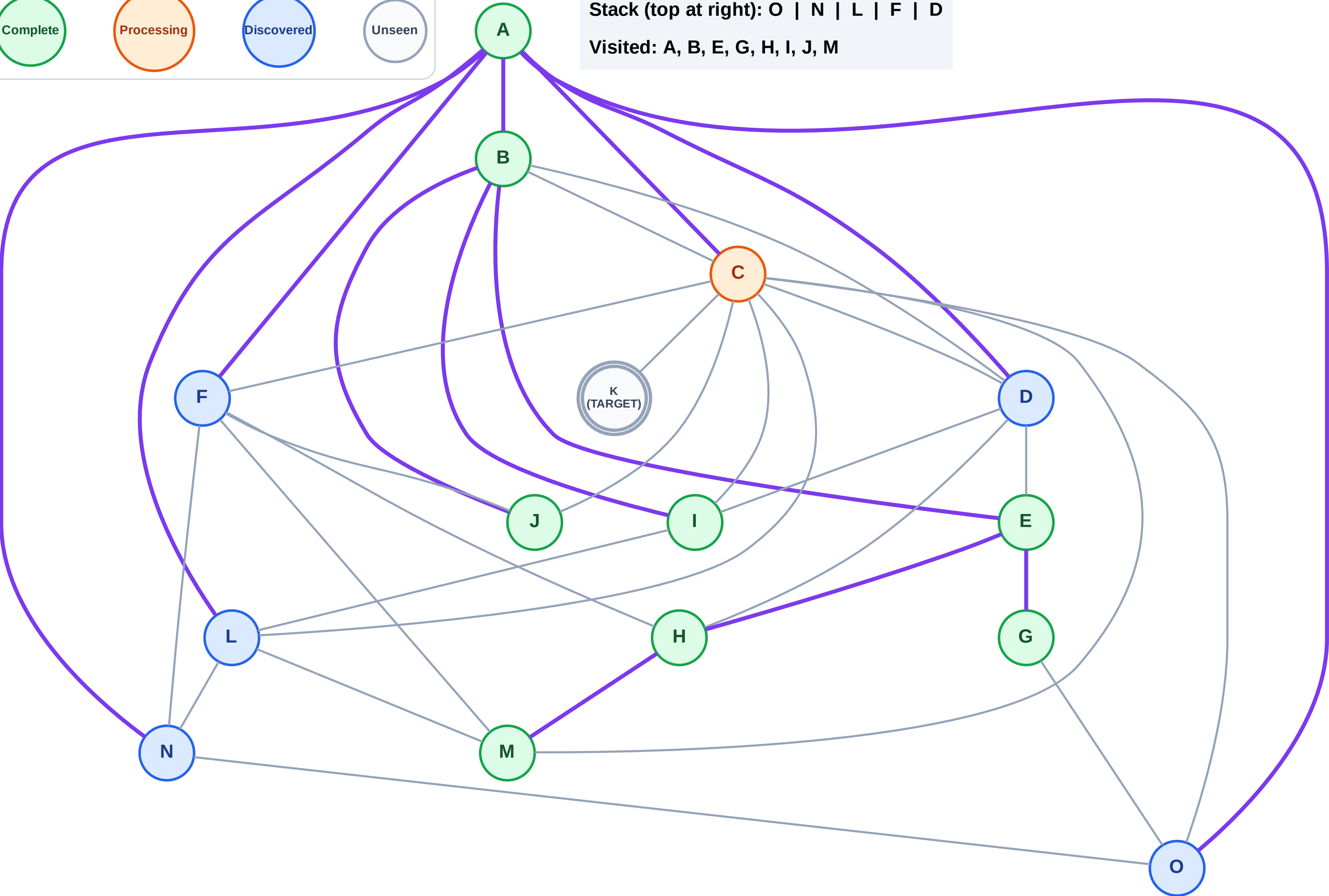
Complete

Processing

Discovered

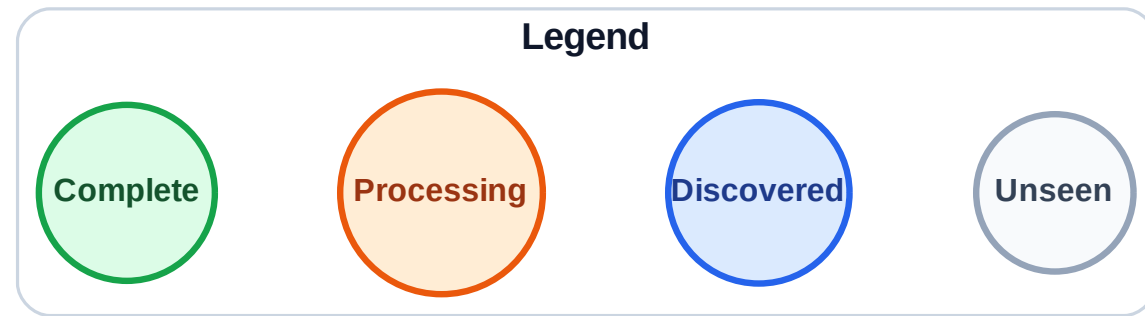
Unseen

Stack (top at right): O | N | L | F | D
Visited: A, B, E, G, H, I, J, M



DFS Traversal

Step 19: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K



Stack (top at right): O | N | L | F | D

Visited: A, B, C, E, G, H, I, J, K, M

